

PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: **GE Varnish for Non-galvanic Flip Chip Bonding**

Requestor: **Rachel Gruenke**

Date: **11/18/2020**

Non-SNF staff, please complete the following:

Badger ID: **rgruenke**

Company name/PI: **Amir Safavi-Naeini**

Send completed form to: snf-promcommittee@lists.stanford.edu

I've highlighted my changes to the original PROM request in YELLOW.

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the [Stanford Chemical Storage Groups](#) to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

1. **GE Varnish**

2. **Vendor/manufacturer info:** address and phone number, website URL.

1. **GE Varnish**

1. Vendor website (Lakeshore Cryo): <https://www.lakeshore.com/products/categories/overview/temperature-products/cryogenic-accessories/varnish>
2. Manufacturer website (von Roll USA): https://www.lakeshore.com/docs/default-source/product-downloads/safety-data-sheets/f068-00-00.pdf?sfvrsn=dd43348f_1 [SDS link]

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

I am developing a non-galvanic flip chip bonding method, using **varnish** as a bonding agent. This process gives us more control in our bonding separation than indium bump-type bonding while keeping our circuit parameters unchanged. The process requires less than **0.5 microliter drop of varnish to be applied onto the bonding seam of the two chips** with a needle or pipette. We have tried using SPR 220-7 photoresist, but it is too brittle to be thermal cycled in a

dilution refrigerator; our SPR 220-7 between the two chips fractures during a second thermal cycling to 4K, and we need our chips to thermal cycle at minimum 3 times (but it could certainly be more based on experimental needs). We have also tried Stycast 2850 epoxy + catalyst 24LV, but found this is too viscous to produce the correct gap size between our two chips. To solve this problem, I have thermal cycled old broken chips (Al on Si) bonded with GE varnish. It survived 3 thermal cycles, with no signs of brittleness and correct gap size. Thus, we have the data to demonstrate that we need this agent to allow us to continue with non-galvanic flip chip bonding.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that if the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

Step 0: Before the flip chip process, the chips are fabricated with Al spacers on the chip, which act as a bond height separator, as well as a barrier for the glue. To ensure that the glue stays far from the bottom chip edge, so that it can't flow onto the Finetech bonder surface when bonding, bond seam must be >1mm from the edge of the bottom chip.

Step 1: Enable the Flip Chip Bonder tool (contaminated category) in ExFab. Install our LINQS lab only heated submount and top arm vacuum chuck. Align our top chip and prepare the recipe.

Step 2: Place the bottom chip on the Finetech Lambda vacuum surface. Align the chips using the Finetech bonder tool, bring the flip chip bonder arm down to bond the chips. Using a **25 gauge needle and syringe**, place 2 small drops of GE Varnish on the bottom chip, one on each bonding seam.

Step 3: Enable temperature profile, and complete the non-galvanic bonding. Remove the chip and disable the flip chip bonder tool after the temperature profile completes.

Step 4: Dispose of the used needle in the sharps container.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

GE varnish solution is in a liquid state. It is mixed with ethanol at a 90:10 ratio by weight (ethanol:VGE) .

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6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). [Storage groups](#) A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while [storage groups](#) C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

[This mixture will be stored in our own lab, and carried in with a secure container when needed.](#)

7. **Disposal:** How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the [SNF Lab members Safety Manual](#) for the different methods of waste disposal that are available in the lab.

[All of these products are solvent based, so they will be disposed of with solvent waste. The syringe and needle used for application will be place in a disposable bag and go in the box for solvent contaminated sharps. The foil used under the chip during application that may collect some of the materials will go in a disposable bag and go in solvent waste.](#)

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: